

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Lead the design and development of backend **microservices** and **RESTful APIs** in **Java** and **Python**, translating business and technical requirements into scalable application designs for a production system handling millions of daily transactions at **99%+ uptime**.
- Built and integrated **AI tooling** (**OpenAI Agents SDK**, **MCP** tool-use, **LangGraph**, **CrewAI**, **AutoGen**) into engineering workflows, applying hands-on experience with **LLMs**, **NLP**, and **agentic AI** to automate business processes across the team.
- Supported cloud and container-based deployment on **AWS** and **GCP** (**Docker**, **Kubernetes**) with multi-stage **CI/CD** pipelines, automating build, test, and progressive deployment to reduce manual release overhead.
- Reviewed code, guided technical decisions, and troubleshoot performance and defect issues across the codebase; mentored engineers on **code review** standards and cloud-native architectural best practices.
- Collaborated with cross-functional teams and project leadership to manage deliverables and track progress against delivery and quality standards.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Led the design and development of 25+ production **Java**, **J2EE**, and **Spring Boot** **microservices** exposing **RESTful APIs** for financial transaction processing, serving 1M+ users at **99.9% uptime** over 4 years on **AWS** (**EC2**, **RDS**, **SQS**, **Lambda**).
- Designed event-driven, **Kafka**-based asynchronous communication between microservices for real-time financial data processing, ensuring data consistency and fault tolerance across distributed services.
- Standardized **REST API** contracts, exception handling, and **Spring Security/OAuth**-aligned authentication patterns across 25+ services, reducing production escalations **60%** through consistent retry, timeout, and failure-handling design.
- Applied Core **Java** (**Java 8+**) concepts, including **Collections**, **Streams**, **Lambdas**, and strong object-oriented design, to build and refactor high-throughput **SQL**-backed services; wrote **JUnit**/**Mockito** test suites and enforced code coverage standards.
- Led architecture decisions and code-review sessions, mentored engineers on system design, reliability, and secure API design, and established code review standards adopted across the team.
- Partnered directly with enterprise client stakeholders to translate ambiguous business requirements into technical designs, using **Agile** ceremonies and estimation to manage delivery and track business impact.

PROJECTS

LangFetch

tinyurl.com/LangFetch

- Agent decomposing natural-language queries into dependency-ordered **SQL** subqueries with **MCP**-compatible tool-use; **FastAPI** backend deployed as an API server for enterprise data queries.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Full-stack microservice with **FastAPI** backend, **PostgreSQL**, and automated **JUnit**-style test suites wired into **CI/CD**, applying object-oriented design patterns for rubric-based evaluation and regression detection.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- Multi-agent system automating production incident diagnosis via tool-use chains, applying **AI/NLP** concepts and object-oriented design to route alerts to specialized sub-agents and synthesize root cause reports.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Java (8+), Python, SQL, TypeScript

Java/Backend: Spring Boot, Spring MVC, Spring Security, J2EE, RESTful APIs, Microservices, OOP Design, JUnit, Mockito, OAuth

Messaging & Data: Kafka, Distributed Messaging, RDBMS/SQL, NoSQL, PostgreSQL, Redis

Cloud & DevOps: AWS (EC2, RDS, SQS, Lambda), GCP, Docker, Kubernetes, Git, Jenkins, GitHub Actions, CI/CD

AI/ML: LLMs, NLP, Machine Learning, Computer Vision, Agentic AI, OpenAI Agents SDK, MCP, LangGraph

CERTIFICATIONS

AWS Certified Developer Associate | Google Cloud Professional Machine Learning Engineer